Intelligence at the Edge: Why TinyML and Why Now

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Thanks

The slides are based on Pietro Manzoni's course "IoT and edge computing: stories of a necessary relationship".

Pietro Manzoni is Professor at Universitat Politècnica de València (UPV).

http://grc.webs.upv.es/

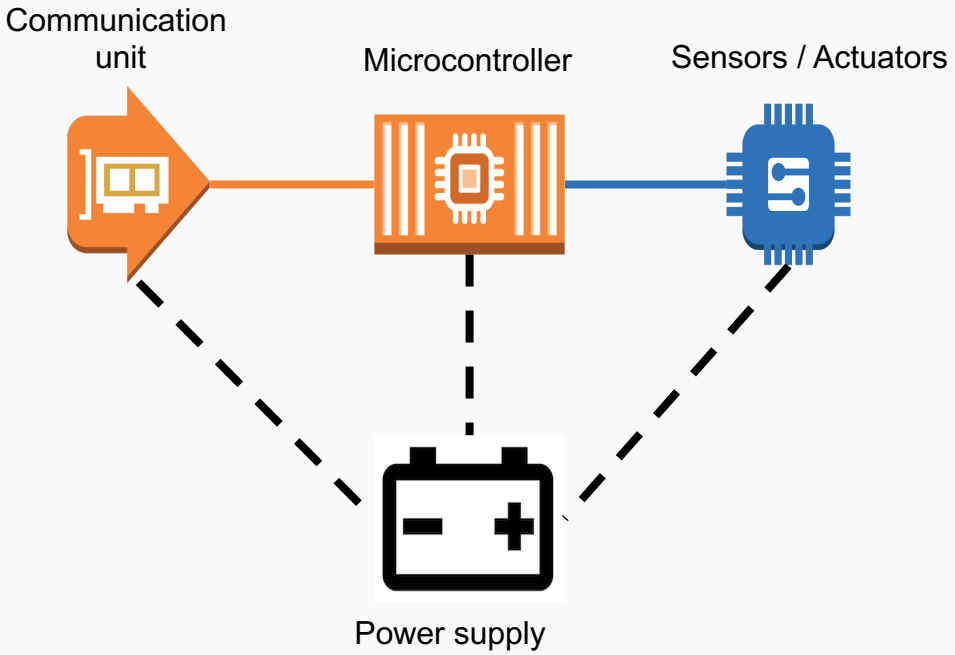
TinyML = IoT + ML

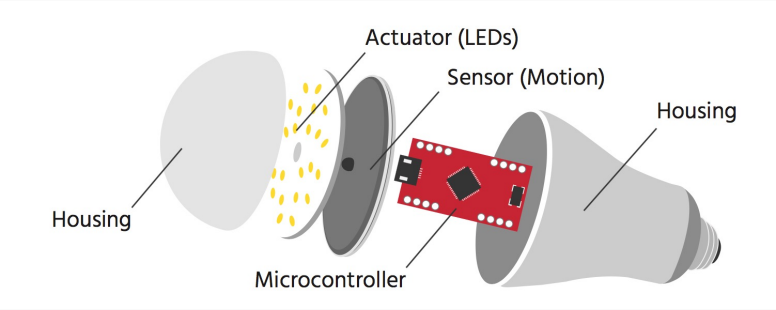
- The **Internet of Things** (IoT) refers to a system of interrelated, Internet-connected objects ("*things*"), that can collect and transfer data.

- **Machine Learning** is typically associated with computationally heavy cloud-based solutions with relatively high latencies, high power consumption, and the need for high bandwidths links.

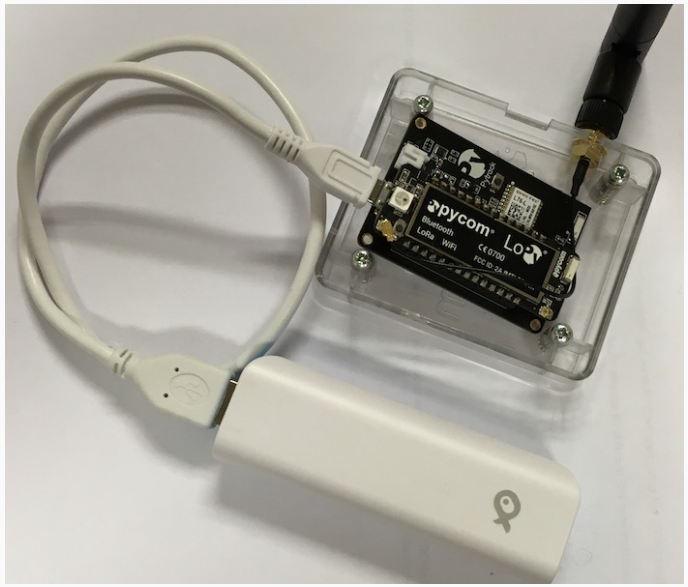
- **With TinyML, IoT and ML come together by shrinking deep learning networks to fit on tiny hardware, thus requiring low energy, low bandwidth links, and reduced latencies.**

Things



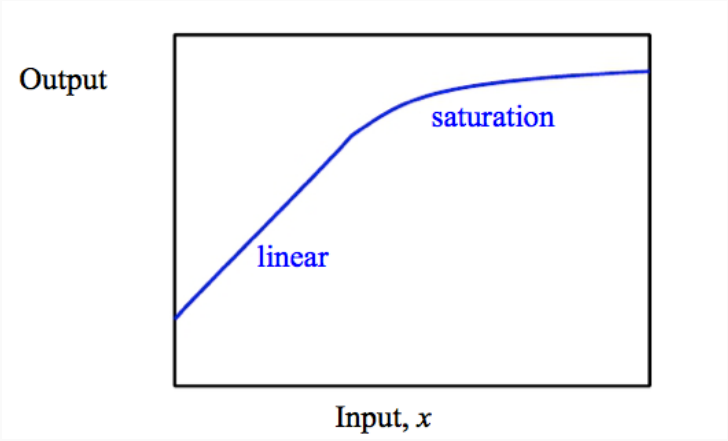


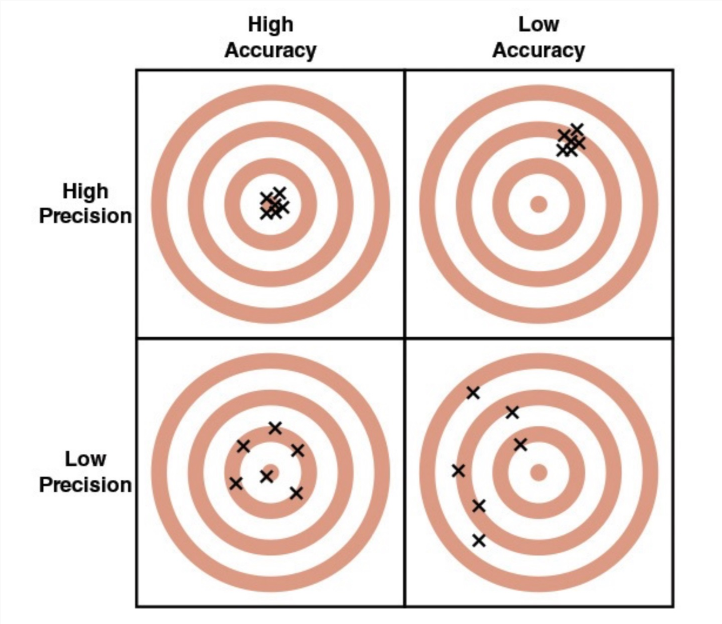
A Reference Guide to the Internet of Things Copyright © 2017 Bridgera LLC, RIoT



Sensors

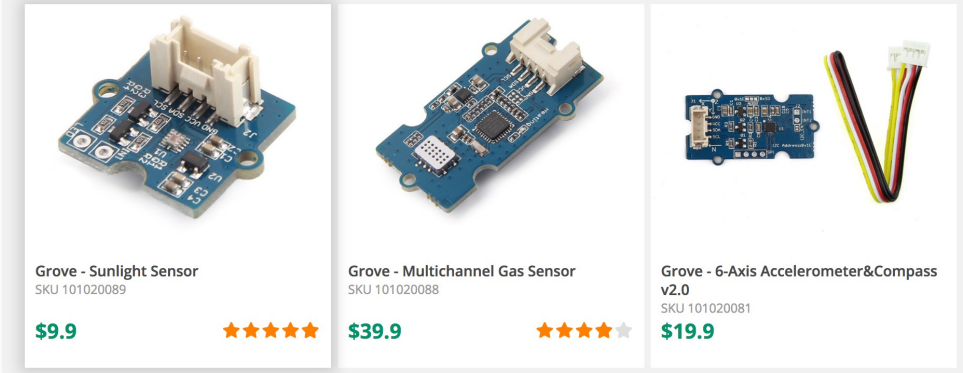
- **Saturation**:
  - The output becomes flat at some levels of input
- **Resolution**:
  - Describe the smallest increment of stimulus
  - Outputs change in small steps
- **Accuracy**:
  - Describes how close the measurement is to the true value
- **Precision** or Repeatability:
  - Describes the difference in results when measurements are taken repeatedly under the same conditions

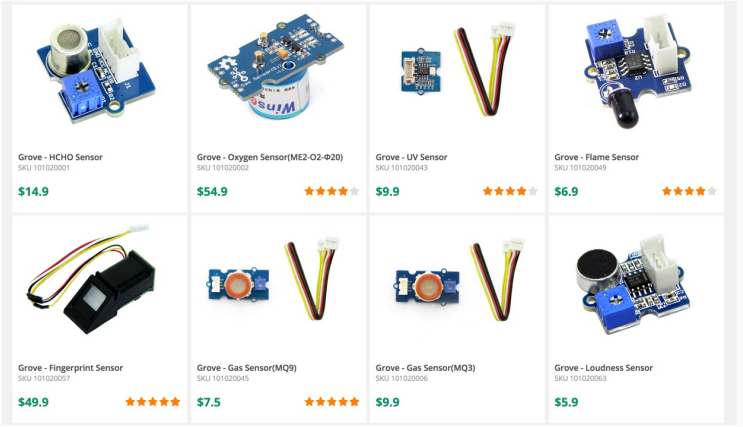




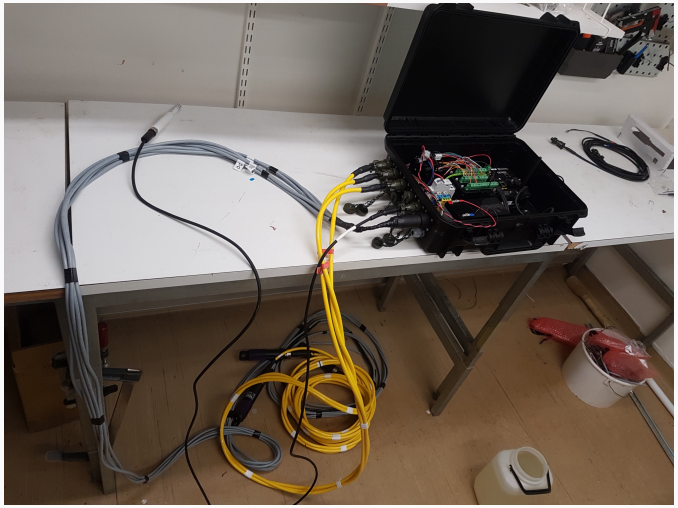
Classic Sensors

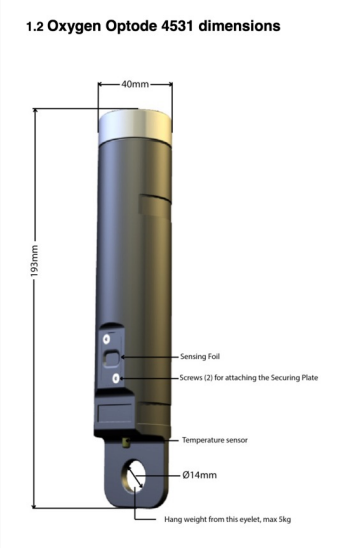
cheap…





expensive…





Parameter	Output	Default range[2)]	Calibrated range	Accuracy	Resolution
Oxygen Concentration	0 - 5V	0 to 800µM	0 to 500µM	<8µM or 5% whichever is greater	< 1µM
	4 - 20mA	0 to 800µM	0 to 500µM	<9µM or 5.2% whichever is greater	< 1µM
Oxygen Saturation	0 - 5V	0 – 200%	0 - 120%	<5 %	<0.4%
	4 - 20mA	0 – 200%	0 - 120%	<5.2 %	<0.4%
Temperature	0 - 5V	-5 to + 35°C	0 - 36°C	±0.1°C	±0.01°C
	4 - 20mA	-5 to + 35°C	0 - 36°C	±0.15°C	±0.02°C

Power sources

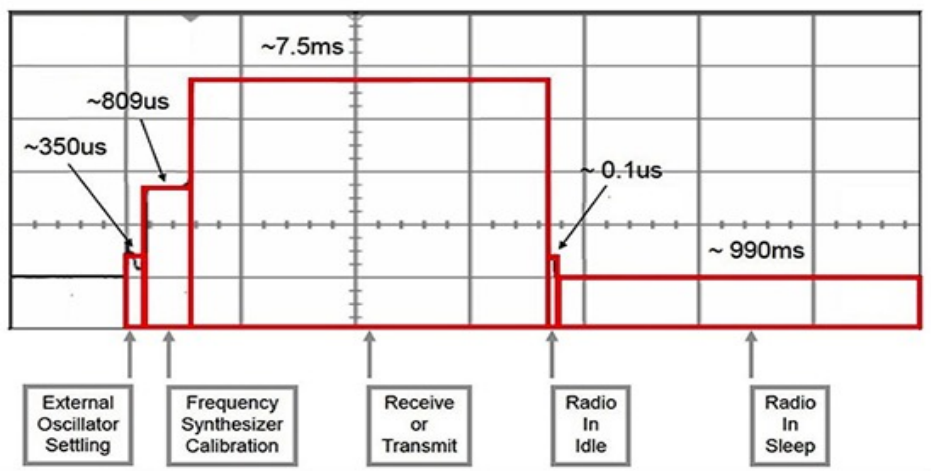
- Today, the **most common power source is a battery**, but there are several other possibilities for power, such as **solar cells, piezoelectricity, radio-transmitted energy, and other forms of power scavenging.**
- Rechargeable batteries are not particularly well-suited to smart objects.
  - Instead of using rechargeable batteries, battery-equipped smart objects are typically designed so a single battery should last the entire lifetime of the smart object.



Lithium Ion (Li-ion)
Lithium-Ion Polymer (Li-Po)

Communication Device

- **Of the hardware components of a “thing”, the radio is usually the most power-consuming component.**
  - Compared to the power consumption of the microcontroller or the sensors, the radio transceiver often uses ten times as much power.
- For low-power radios, only a small portion of the power consumption is used to send the radio signal into the air.
  - ➔ listening is as power consuming as sending.



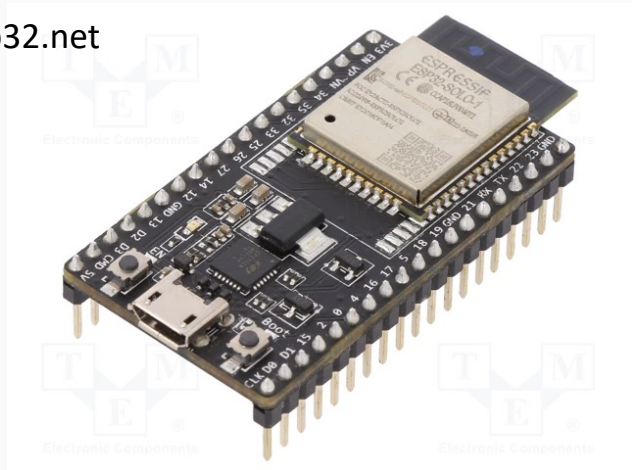
Microcontroller: ESP32

- The ESP32 series employs a Tensilica Xtensa LX6 microprocessor in both dual-core and single-core variations. ESP32 is created and developed by Espressif System

- CPU: Xtensa dual-core (or single-core) 32-bit LX6 microprocessor, operating at 160 or 240 MHz and performing at up to 600 DMIPS
- Memory: 520 KiB SRAM
- Wireless connectivity:
  - Wi-Fi: 802.11 b/g/n
  - Bluetooth: v4.2 BR/EDR and BLE
- Peripheral interfaces:
  - 12-bit SAR ADC up to 18 channels
  - 2 × 8-bit DACs
  - 10 × touch sensors (capacitive sensing GPIOs)
  - …

https://www.espressif.com/en/products/socs/esp32

http://esp32.net



Mode	Min	Typ	Max	Unit
Transmit 802.11b, DSSS 1 Mbps, POUT = +19.5 dBm	-	240	-	mA
Transmit 802.11g, OFDM 54 Mbps, POUT = +16 dBm	-	190	-	mA
Transmit 802.11n, OFDM MCS7, POUT = +14 dBm	-	180	-	mA
Receive 802.11b/g/n	-	95 ~ 100	-	mA
Transmit BT/BLE, POUT = 0 dBm	-	130	-	mA
Receive BT/BLE	-	95 ~ 100	-	mA

Power mode	Description			Power consumption
Modem-sleep	The CPU is powered on.	240 MHz *	Dual-core chip(s)	30 mA ~ 68 mA
			Single-core chip(s)	N/A
		160 MHz *	Dual-core chip(s)	27 mA ~ 44 mA
			Single-core chip(s)	27 mA ~ 34 mA
		Normal speed: 80 MHz	Dual-core chip(s)	20 mA ~ 31 mA
			Single-core chip(s)	20 mA ~ 25 mA
Light-sleep	-			0.8 mA
Deep-sleep	The ULP coprocessor is powered on.			150 μA
	ULP sensor-monitored pattern			100 μA @1% duty
	RTC timer + RTC memory			10 μA
Hibernation	RTC timer only			5 μA
Power off	CHIP_PU is set to low level, the chip is powered off.			1 μA

No good data left behind

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No Good Data Left Behind

5 Quintillion

bytes of data produced every day by IoT

<1%

of unstructured data is analyzed or used at all

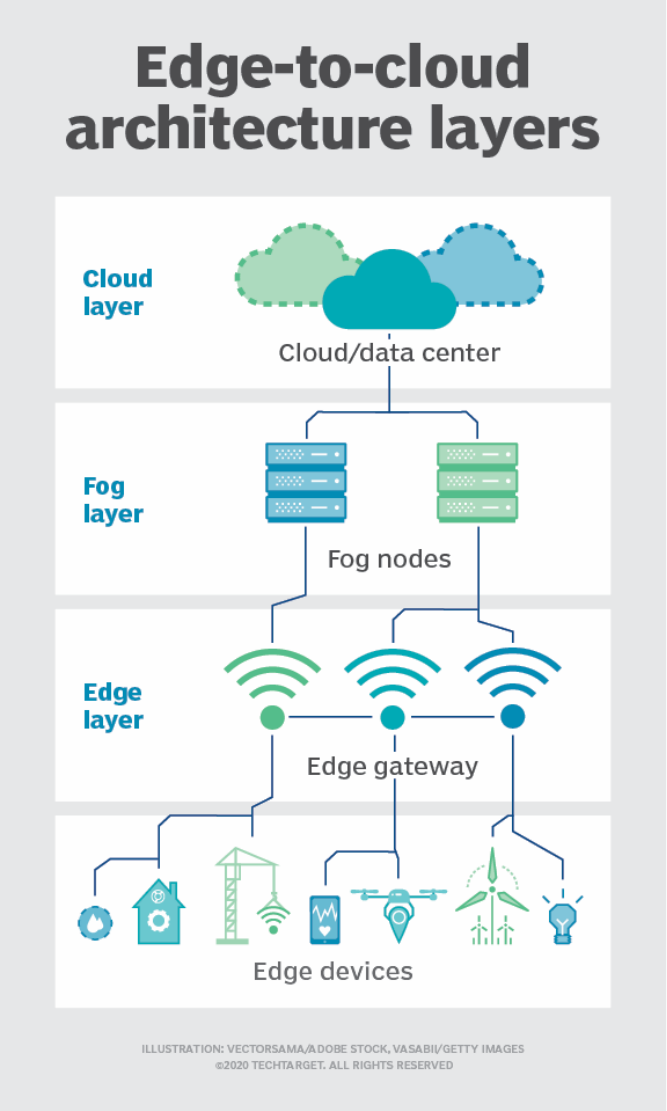
Source: Harvard Business Review, What's Your Data Strategy?, April 18, 2017
Cisco, Internet of Things (IoT) Data Continues to Explode Exponentially. Who Is Using That Data and How?, Feb 5, 2018

What does that all mean?



So… Edge Computing

- A possible definition of edge computing is: “a part of a distributed computing topology in which information processing is located close to the edge — where things and people produce or consume that information.”



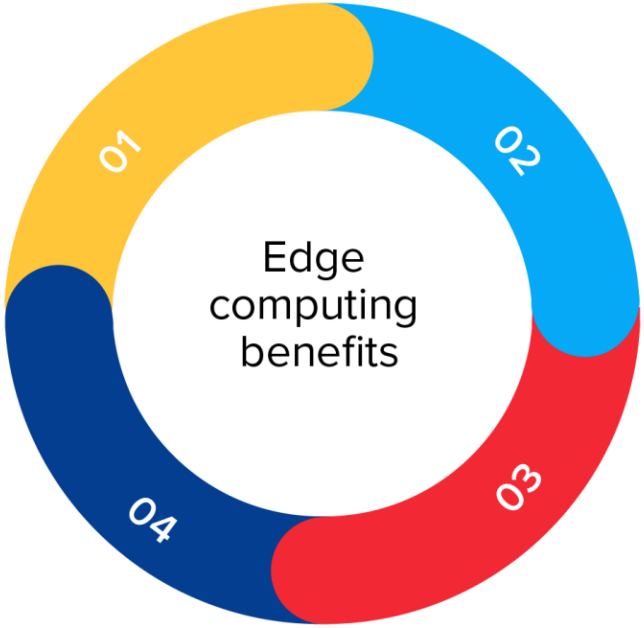
Advantages of Edge Computing… for IoT



Latency
Reduction of latency by processing the data closer to the customer



Security ➔ Privacy
Computing at the edge provides more security than computing at the cloud because it is less vulnerable to numerous variety of threats due to its scope





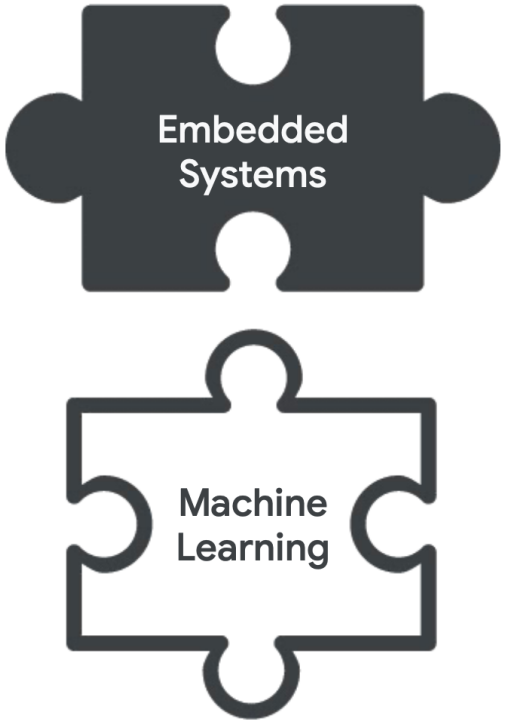
Bandwidth
Sending data from edge to the cloud takes up spectral resources; there's just not enough bandwidth for data transportation



Reliability
By processing data at the edge, you eliminate network reliability problems

https://www.wwt.com/article/show-me-the-money-drive-new-revenue-streams-with-edge-computing

What is TinyML?



Original definition: "…a neural network model that runs at an energy cost of below 1 mW."
© "TinyML" by Pete Warden, Daniel Situnayake

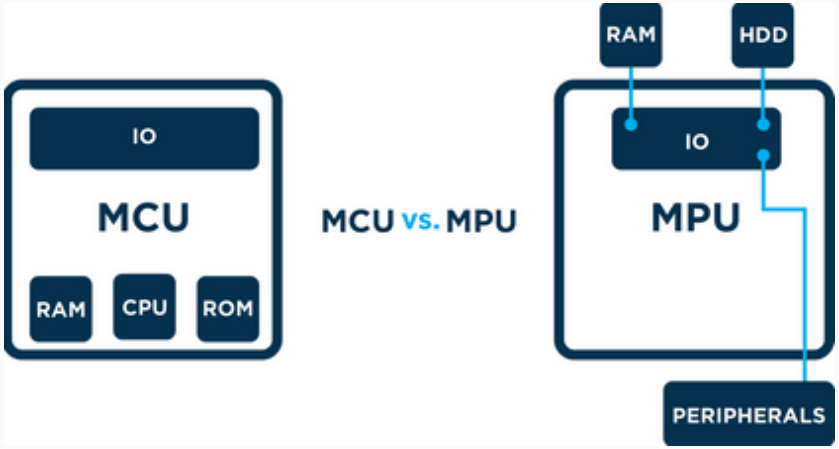


TinyML

"Tiny machine learning (TinyML) is a fast-growing field of machine learning technologies and applications including algorithms, hardware, and software capable of performing on-device sensor data analytics at extremely low power consumption, **typically in the mW range and below**, enabling a variety of always-on ML use-cases on battery-operated devices."

Microcontroller

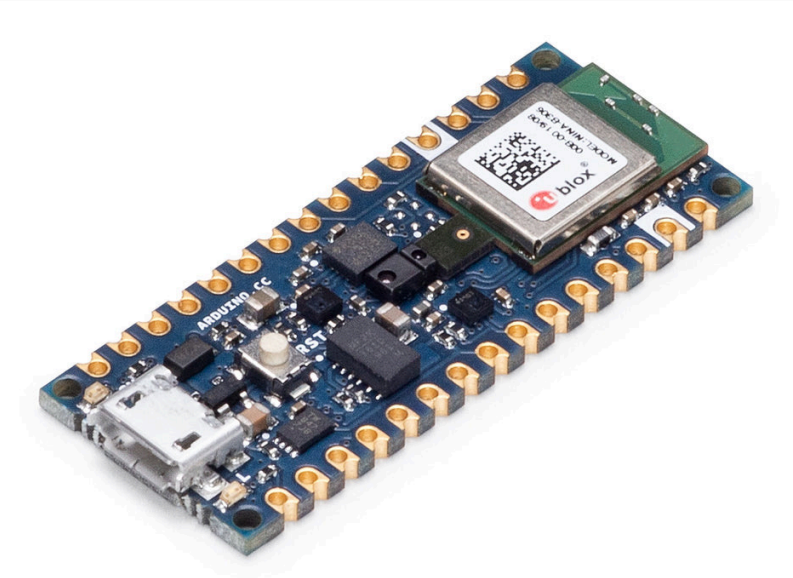
- A microcontroller is a microprocessor with built-in memory, timers, and hardware for connecting external devices such as sensors, actuators, and radio transceivers.
- Typically, a smart object microcontroller has a few kilobytes of on-chip memory and is run at a clock speed of a few megahertz.



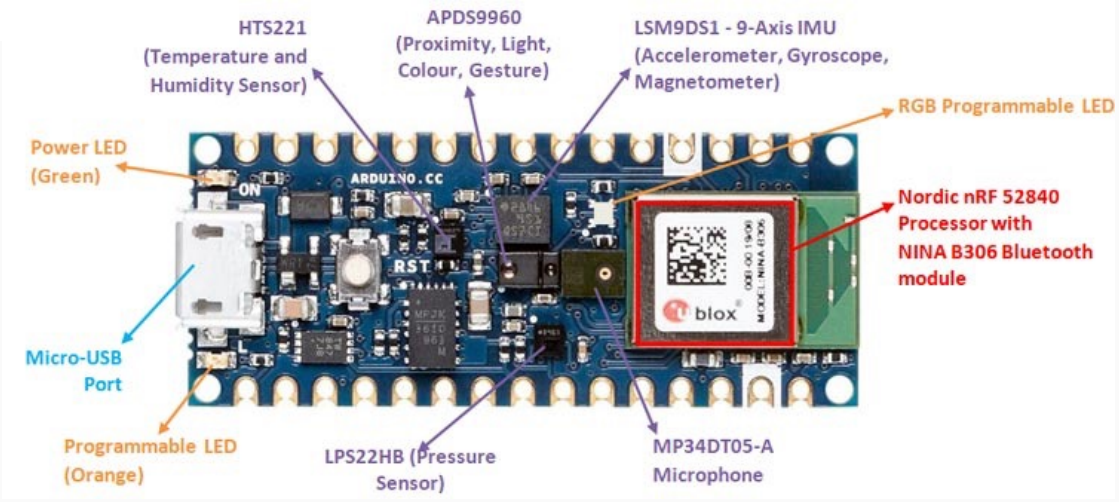
Orders of magnitude

	Microprocessor	>	Microcontroller
Platform			
Compute	1GHz–4GHz	~10X	1MHz–400MHz
Memory	512MB–64GB	~10000X	2KB–512KB
Storage	64GB–4TB	~100000X	32KB–2MB
Power	30W–100W	~1000X	150µW–23.5mW

Most widely used HW for TinyML (currently)



Arduino Nano 33 BLE Sense
64 MHz **Arm® Cortex-M4**F (with FPU)
1 MB Flash + 256 KB RAM
Bluetooth® 5 multiprotocol radio



Or even… https://ouraring.com/



The future of health wrapped around your finger — monitoring your sleep, heart rate, activity, and temperature with personalized insights.

Or even…



Or even…

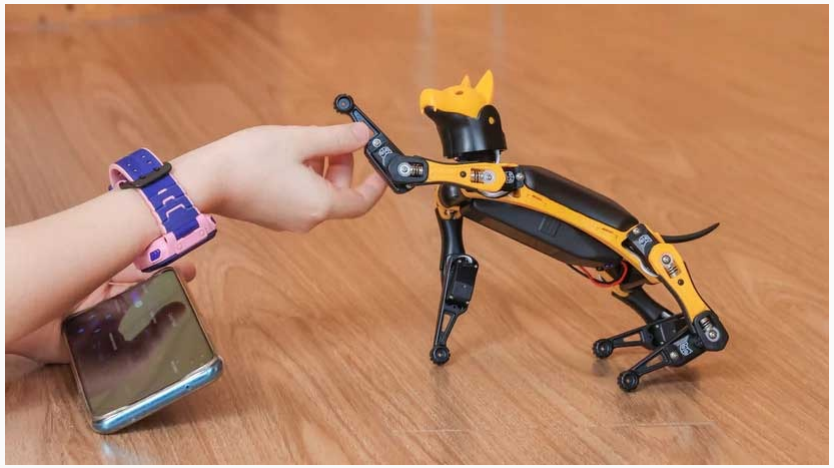
Kicking
Penalty
kicking
Passing
Dribbling
...



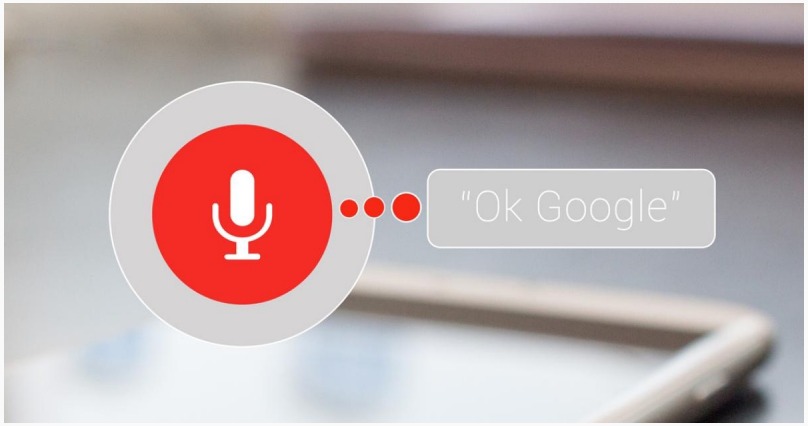
Or even…

Tiny Robot Learning: Challenges and Directions for Machine Learning in Resource-Constrained Robots

Sabrina M. Neuman[1], Brian Plancher[1], Bardienus P. Duisterhof[2], Srivatsan Krishnan[1], Colby Banbury[1], Mark Mazumder[1], Shvetank Prakash[1], Jason Jabbour[3], Aleksandra Faust[4], Guido C.H.E. de Croon[5], and Vijay Janapa Reddi[1]
Harvard University[1], CMU[2], University of Virginia[3], Google Brain[4], Delft University of Technology[5]
{sneuman@seas, brian_plancher@g, srivatsan@g, cbanbury@g, markmazumder@g, sprakash@g, vj@eecs}.harvard.edu, bduister@andrew.cmu.edu, jjj4se@virginia.edu, aleksandra.faust@gmail.com, g.c.h.e.decroon@tudelft.nl



Applications of TinyML: the beginning



But the range of applications is wide…

We help African smallholder farmers adapt to climate change

Help us reach more farmers

Donate ♥ & support farmers →



Plant Village: https://plantvillage.psu.edu

But the range of applications is very wide…



HOW OUR SYSTEM HELPS PRESERVE RAINFORESTS

PREVENT ILLEGAL DEFORESTATION

Our mission is to enable our partners-on-the-ground to protect rainforests. Our system sends real-time alerts for chainsaws, trucks, cars and signs of incursion. Read more

HALT ANIMAL POACHING

We can help stop poaching by providing our partners with real-time data and patterns of activity that allow for targeted protections in key strategic areas. Read more

ENABLE BIO-ACOUSTIC MONITORING

There's more to a rainforest than meets the eye. Our bio-acoustic monitoring capabilities give partners new ways of understanding rainforests. Read more

Rainforest Connection: https://rfcx.org/ecoacoustics

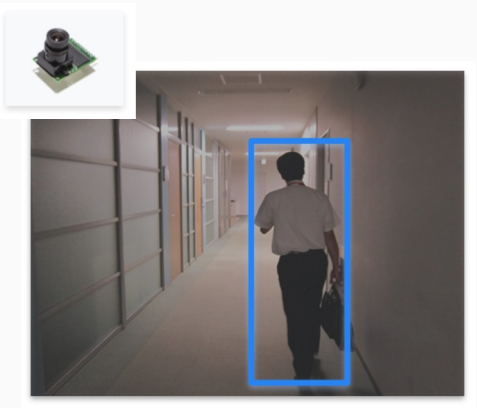
Sound



Vibration



Vision



Sound

Vibration

Vision

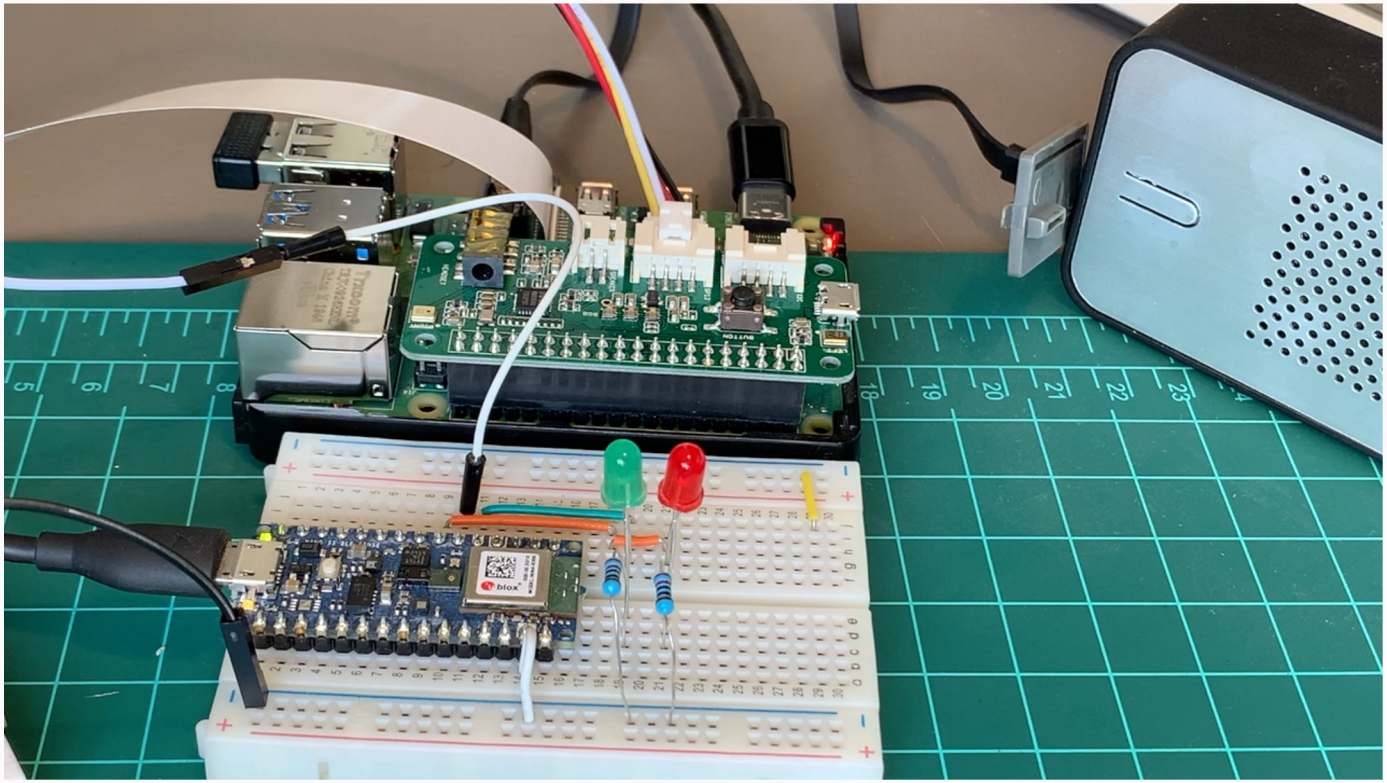


Personal Assistant

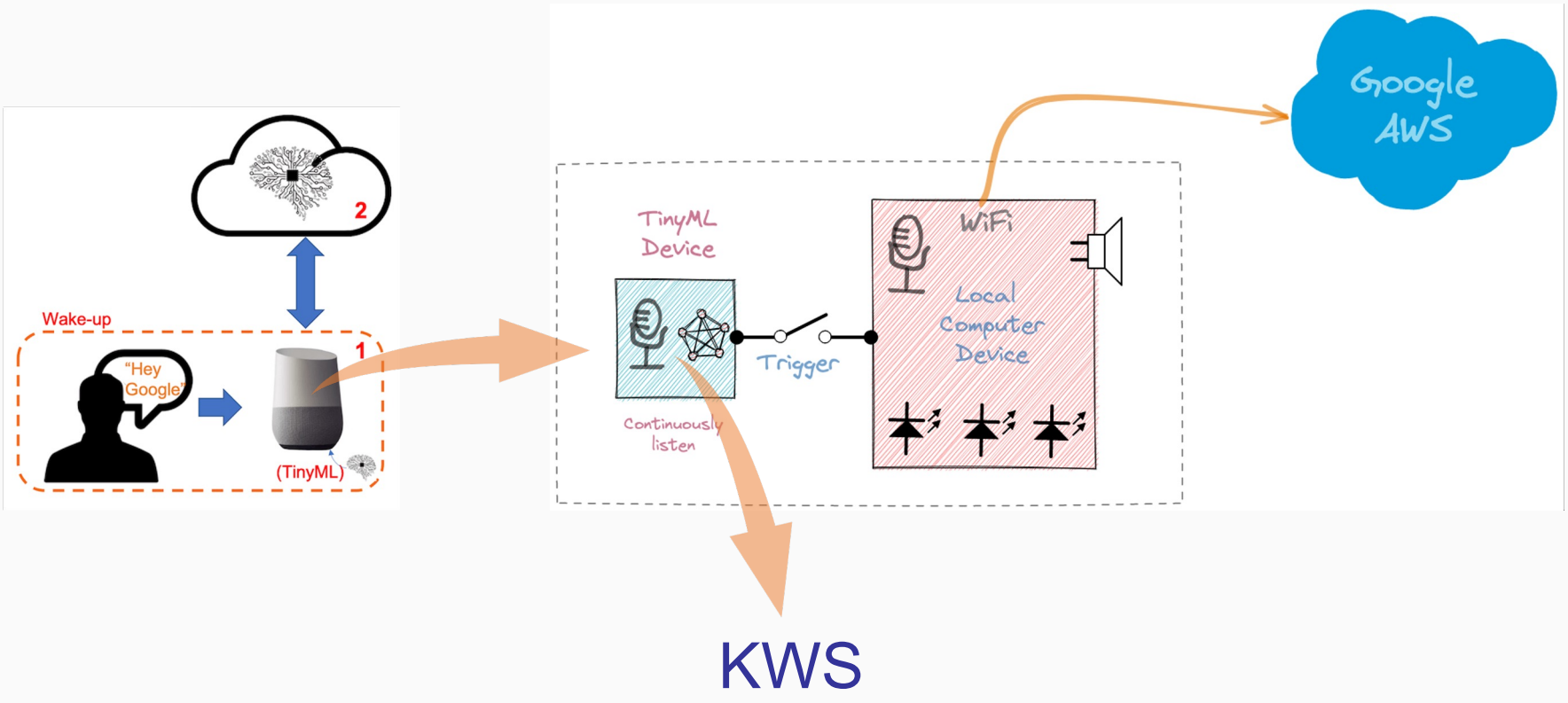




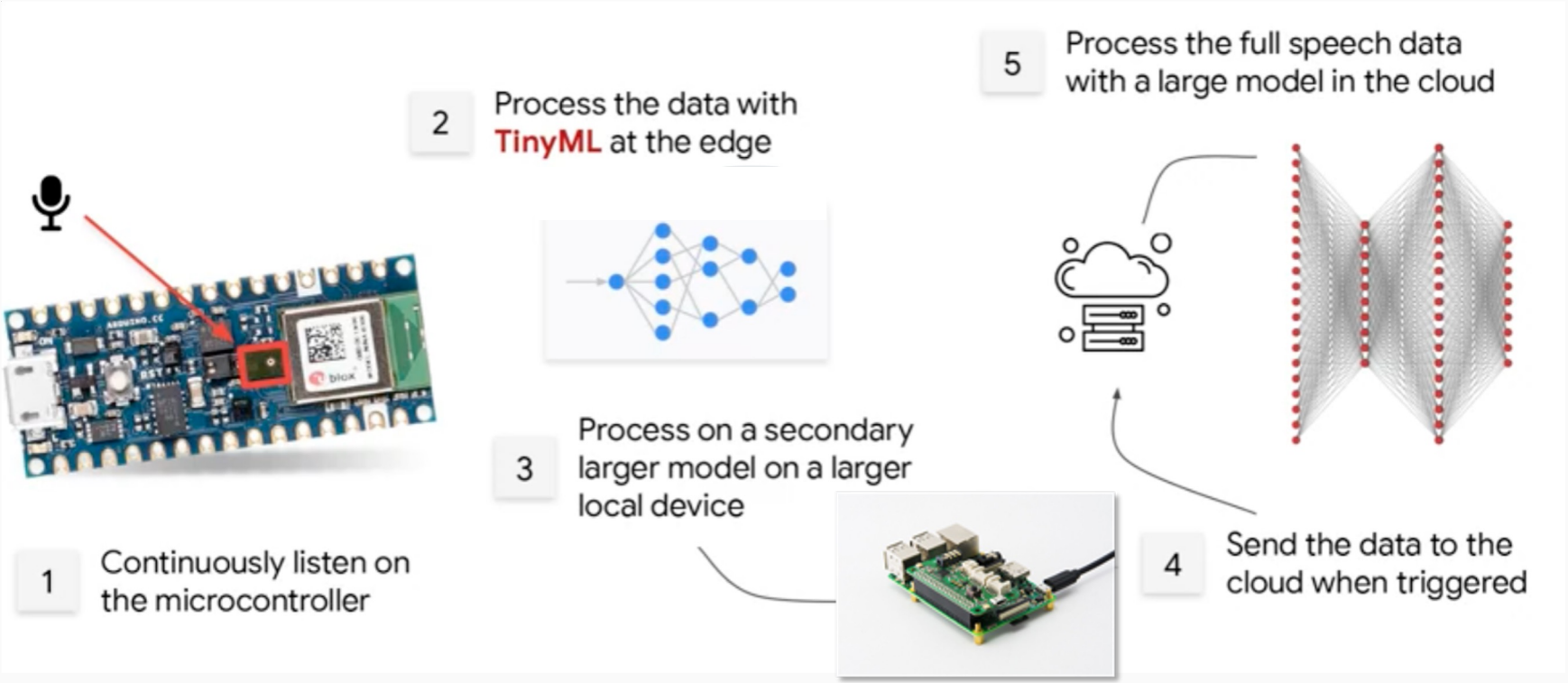
Personal Assistant



Personal Assistant



“Cascade” Detection: multi-stage model





Classifying mosquito wingbeat sound using TinyML

Moez Altayeb
University of Khartoum, Sudan
ICTP, Trieste, Italy
mohedahmed@hotmail.com

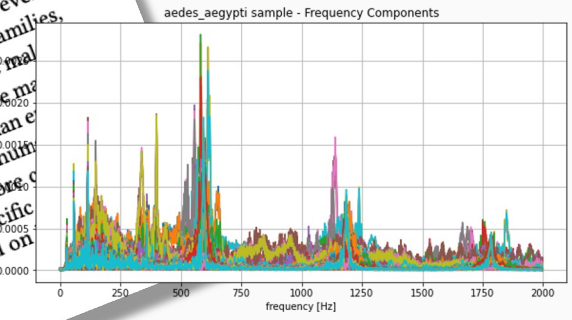
Marcelo Rovai
Universidade Federal de Itajubá
Itajubá, Brazil
rovai@unifei.edu.br

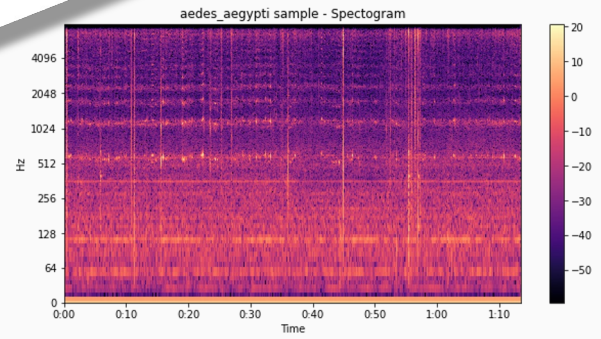
Marco Zennaro
ICTP
Trieste, Italy
mzennaro@ictp.it

ABSTRACT

Every year more than one billion people are infected and more than one million people die from vector-borne diseases including malaria, dengue, zika and chikungunya. Mosquitoes are the best known disease vector and are geographically spread worldwide. It is important to raise awareness of mosquito proliferation by monitoring their incidence, especially in poor regions. Acoustic detection of mosquitoes has been studied for long and ML can be used to automatically identify mosquito species by their wingbeat. We present a prototype solution based on an openly available dataset, on the Edge Impulse platform and on three commercially-available TinyML devices. The proposed solution is low-power, low-cost and can run without human intervention in resource-constrained areas. This insect monitoring system can reach a global scale.

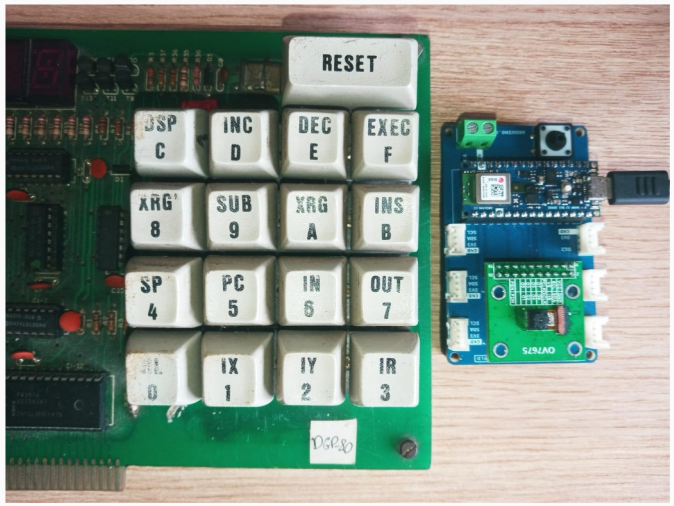
affected. People from poor communities with little access to health care and clean water sources are also at risk. Although anti-malarial drugs exist, there's currently no malaria vaccine. Vector-borne diseases also exacerbate poverty. Illness prevent people from working and supporting themselves and their families, impeding economic development. Countries with intensive ma have much lower income levels than those that don't have ma Countries affected by malaria turn to control rather than e tion. Vector control means decreasing contact between hum disease carriers on an area-by-area basis. It is therefore c be able to detect the presence of mosquitoes in a specific paper presents an approach based on TinyML and on embedded devices.

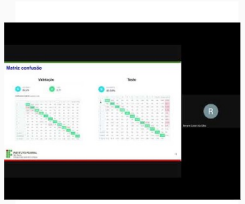




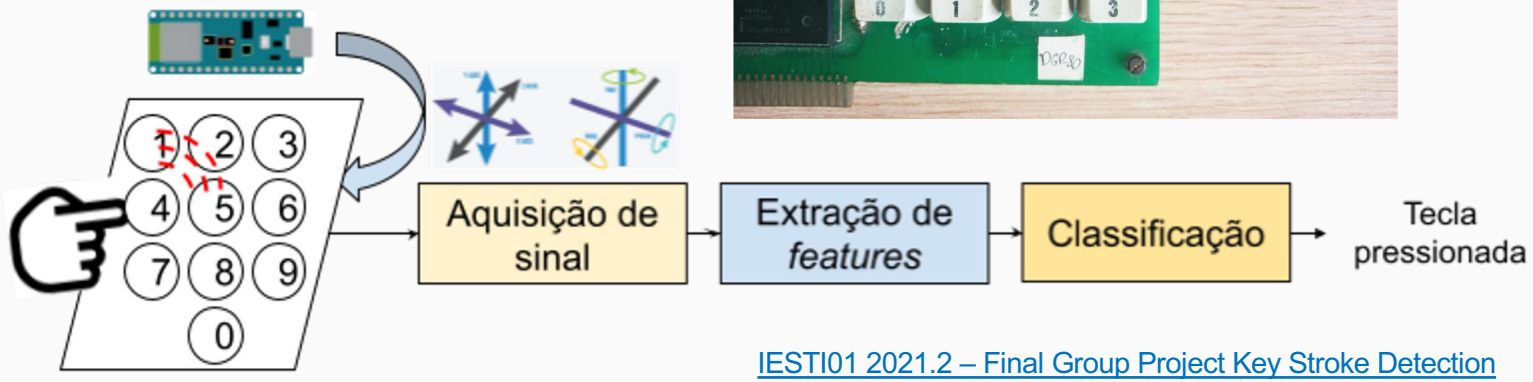
https://www.edgeimpulse.com/blog/the-world-is-all-abuzz-about-tinyml

Keystroke **Sound** Detection





Renam Castro
Professor IFESP



[IESTI01 2021.2 – Final Group Project Key Stroke Detection]

Sound

Vibration

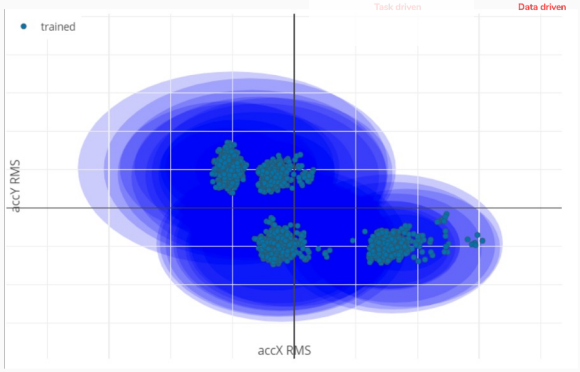


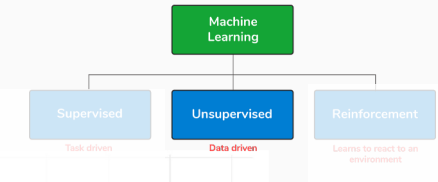
Vision

Industrial – Anomaly Detection

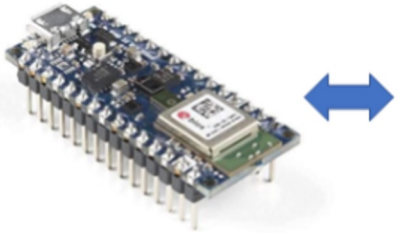


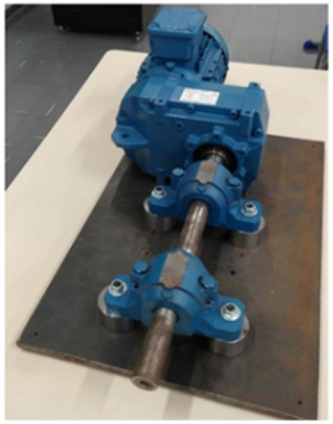














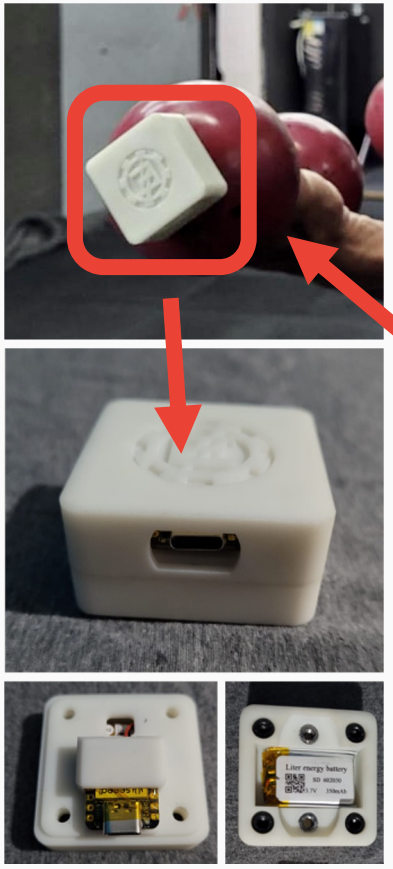


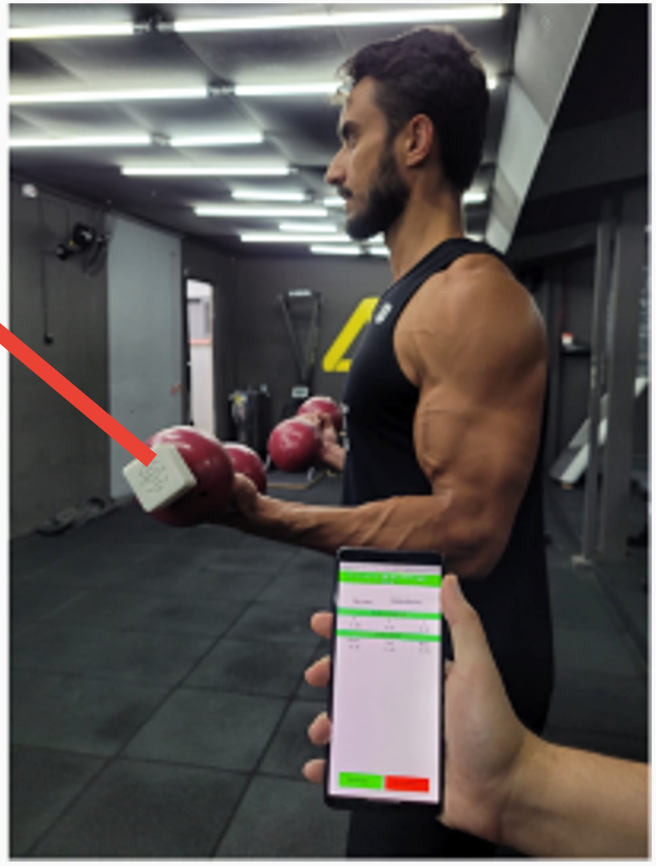


[IESTI01 2021.2 - Final Group Project: Bearing Failure Detection]

Movement Classification





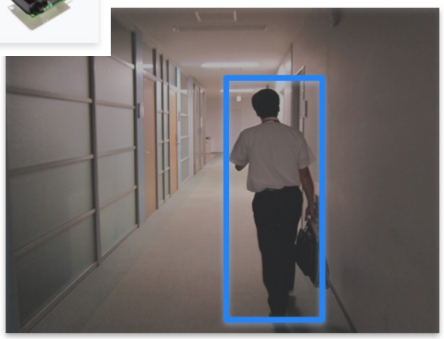


CNPq UNIFEI

Sound

Vibration

Vision



Computer Vision Main Types

Image Classification

(**Multi-Class** Classification)



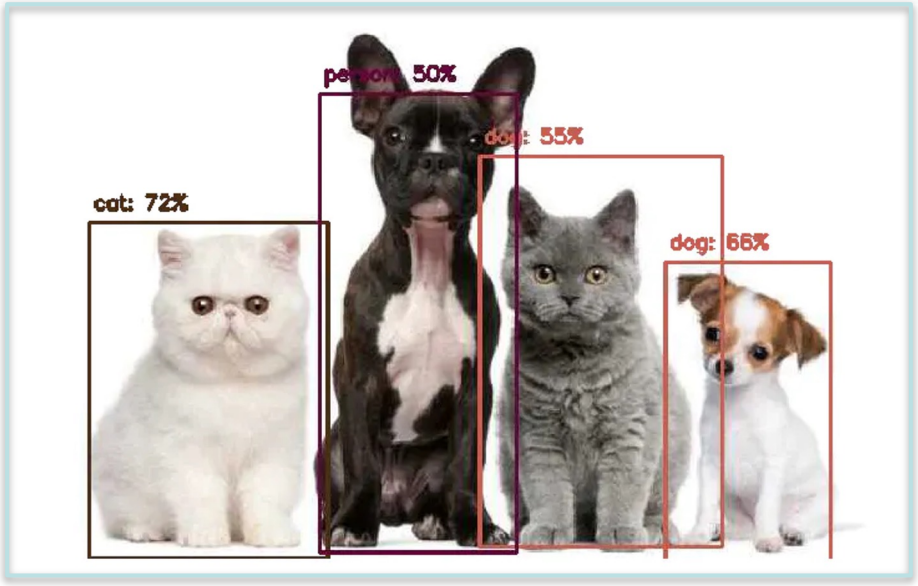
Cat: 70%



Dog: 80%

Object Detection

Multi-Label Classification + Object **Localization**



Computer Vision Main Types

Image Classification
(**Multi-Class** Classification)

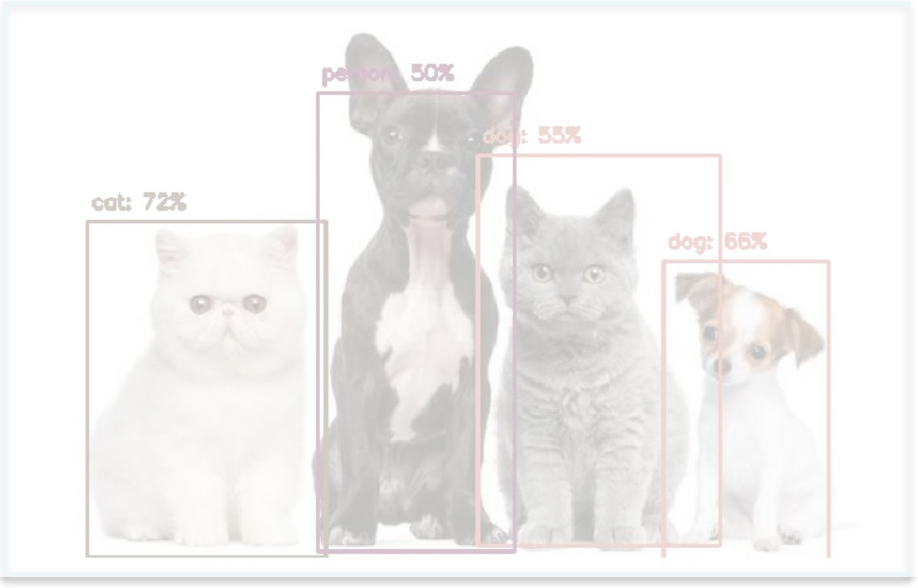


Cat: 70%



Dog: 80%

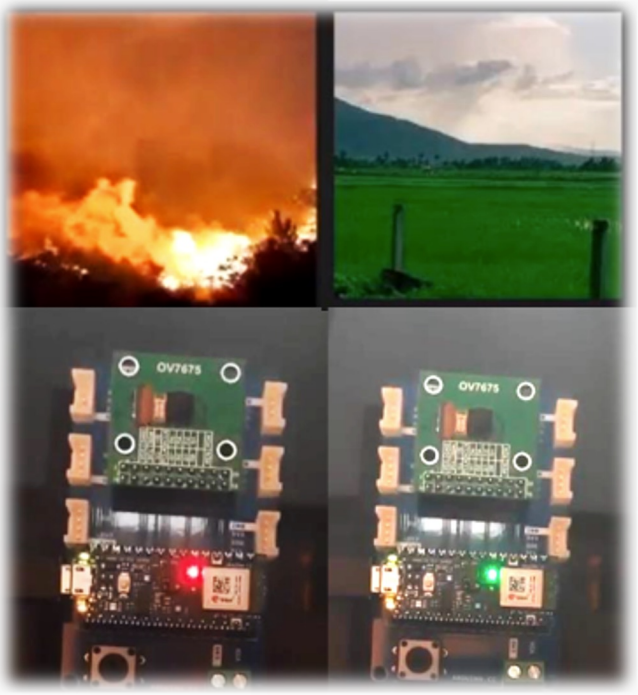
Object Detection
Multi-Label Classification + Object **Localization**



Forest Fire Detection



TinyML Aerial Forest Fire Detection



IESTI01 - Forest Fire Detection – Proof of Concept

Coffee Disease Classification



https://www.hackster.io/Yukio/coffee-disease-classification-with-ml-b0a3fc



João Vitor Yukio Bordin Yamashita
Graduando em Engenharia Eletrônica pela UNIFEI

Computer Vision Main Types

Image Classification
(**Multi-Class** Classification)

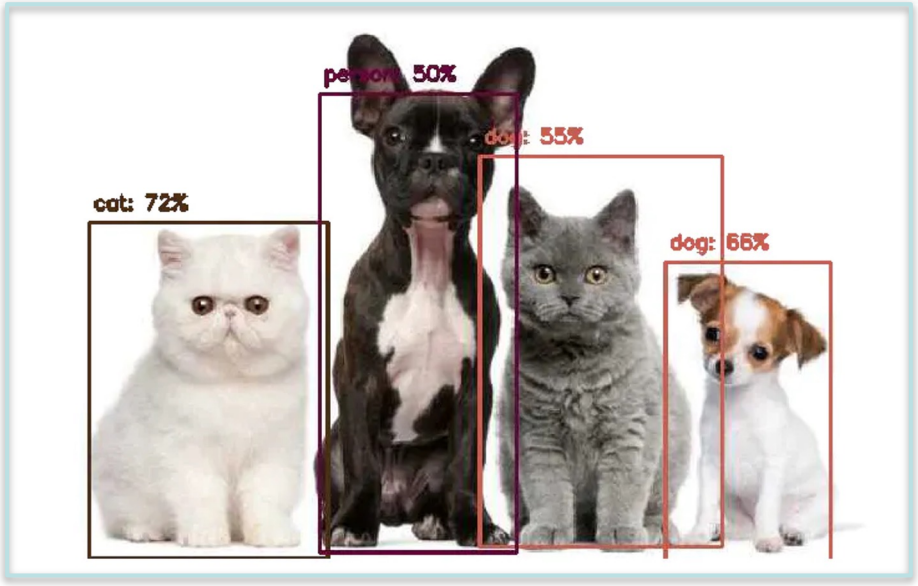


Cat: 70%

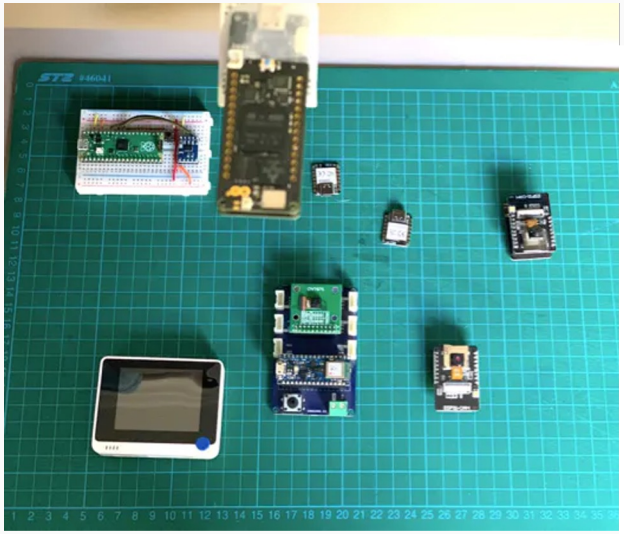


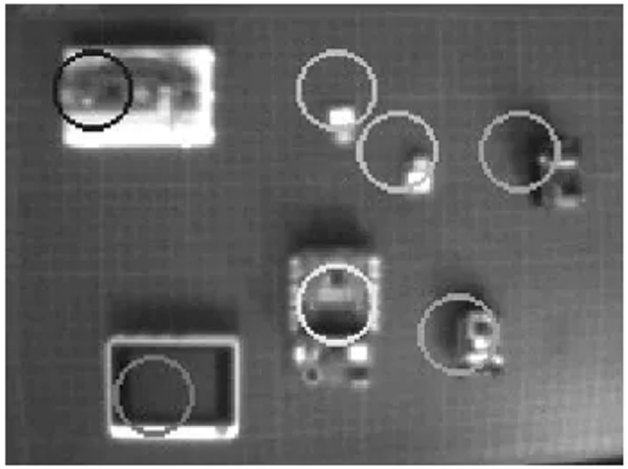
Dog: 80%

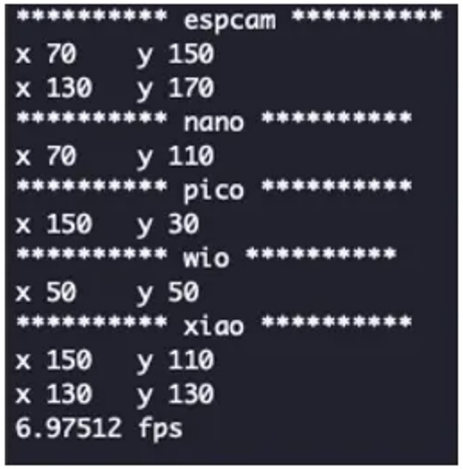
Object Detection
Multi-Label Classification + Object **Localization**



Detecting Objects using TinyML (FOMO)



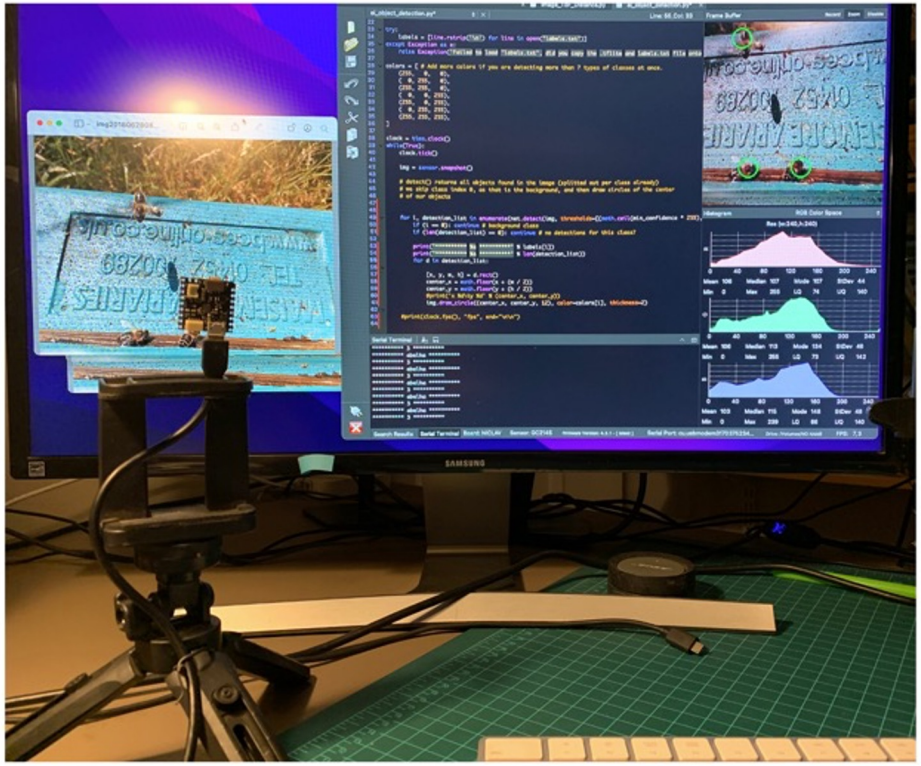




[EdgeAI made simple - Exploring Image Processing (Object Detection) on microcontrollers with Arduino Portenta, Edge Impulse FOMO, and OpenMV]

Detecting Objects using TinyML (FOMO)



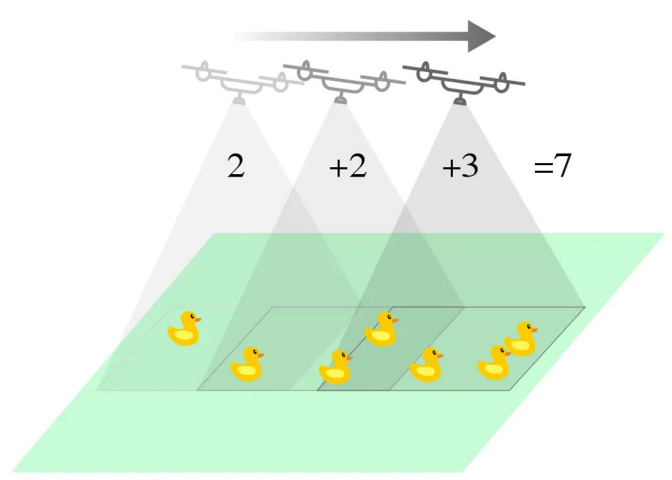


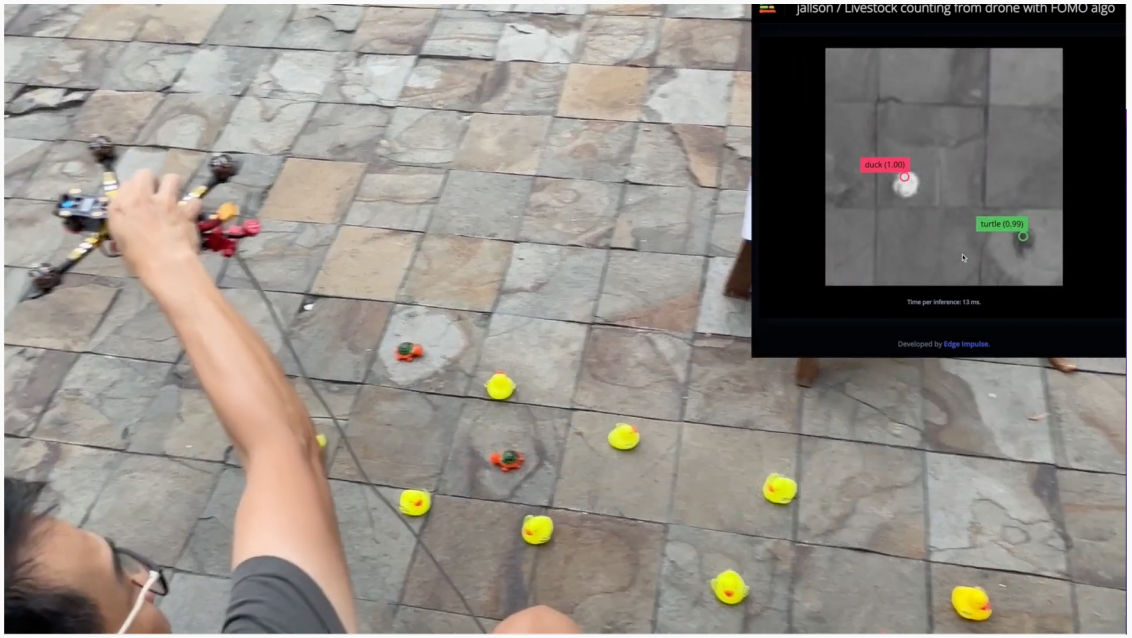
OpenMV

MicroPython

EDGE IMPULSE

Livestock / Wildlife Counting from Drone with FOMO



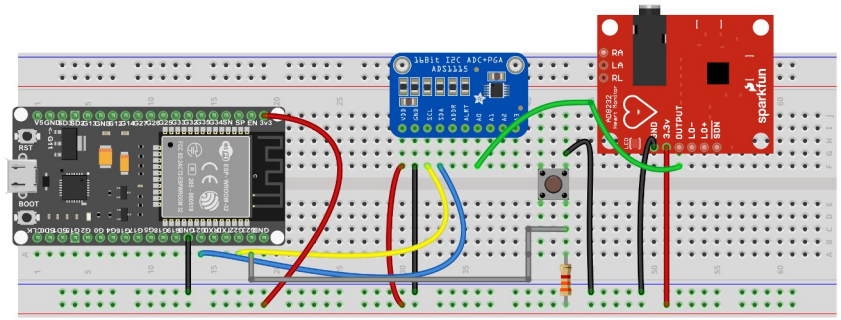


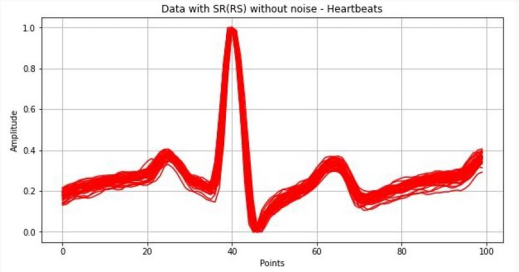
https://www.hackster.io/jallsonsuryo/livestock-wildlife-counting-from-drone-with-fomo-algorithm-a2f734

Other Sensors / MCUs / Models Examples

AD8232 - Single Lead Heart Rate Monitor









Guilherme Silva
Engenheiro - UNIFEI

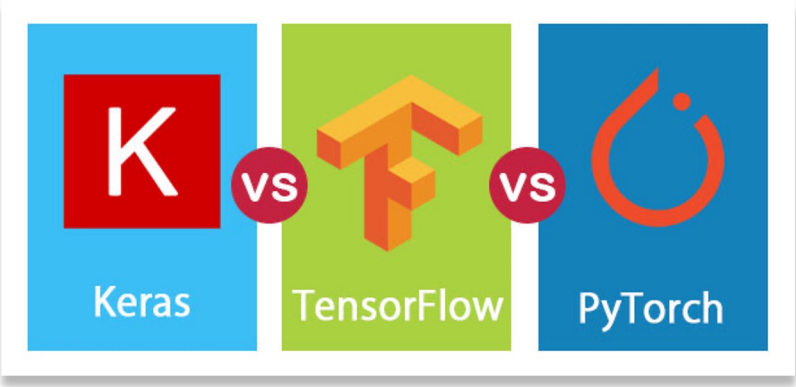
[Atrial Fibrillation Detection on ECG using TinyML
Silva et al. UNIFEI 2021]

Training and Inference

- **Machine Learning** occurs in two stages – learning and inferencing. At present, TinyML only handles inferencing
- Training refers to the process of creating a machine learning algorithm. Training involves using a machine learning framework and a training dataset.
  - IoT data provides a source of training data that data scientists and engineers can use to train machine learning models.
- Inference refers to the process of using a trained machine-learning algorithm to make a prediction.
  - IoT data can be used as the input to a trained machine learning model, enabling predictions that can guide decision logic on the device, at the edge.

About the available software frameworks

- The three main frameworks which are available as an open-source library are opted by data scientist in deep learning are TensorFlow, Keras, and PyTorch.
  - **TensorFlow** is used to perform multiple tasks in data flow programming and machine learning applications.
  - **Keras** is a neural network library scripted in Python and executes on the top layer of TensorFlow.
  - **PyTorch** is a machine learning library that is mainly used in natural language processing.



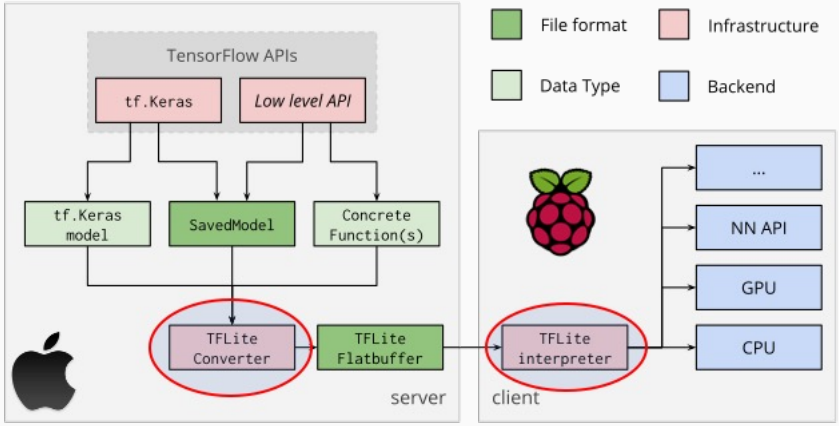
TensorFlow Lite for microcontrolers

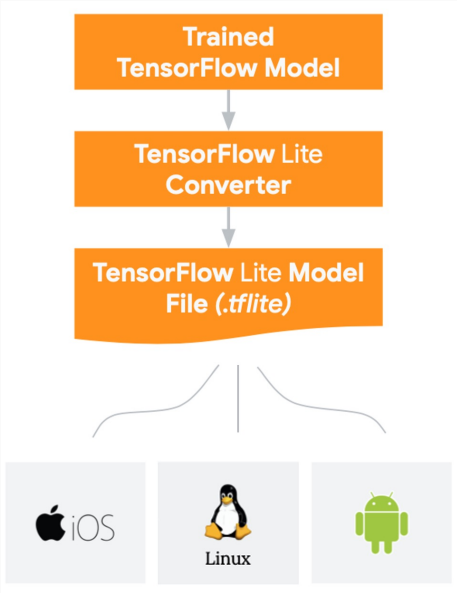


- https://www.tensorflow.org/lite/microcontrollers
- TensorFlow is Google's open-source machine learning framework which helps in developing machine learning models quickly.
- For TinyML, there is TensorFlow Lite Micro, which is a specialised version of TensorFlow for microcontrollers.
  - TFLite Micro is written in C++ 11 and requires a 32-bit platform and is mostly compatible with **Arm Cortex-M** Series processors.
  - An ESP32 port is available as well
- TFLite Micro allows you to easily compress regular TensorFlow models into just a few kilobytes and comes with numerous example models.

TensorFlow Lite (TFLite)

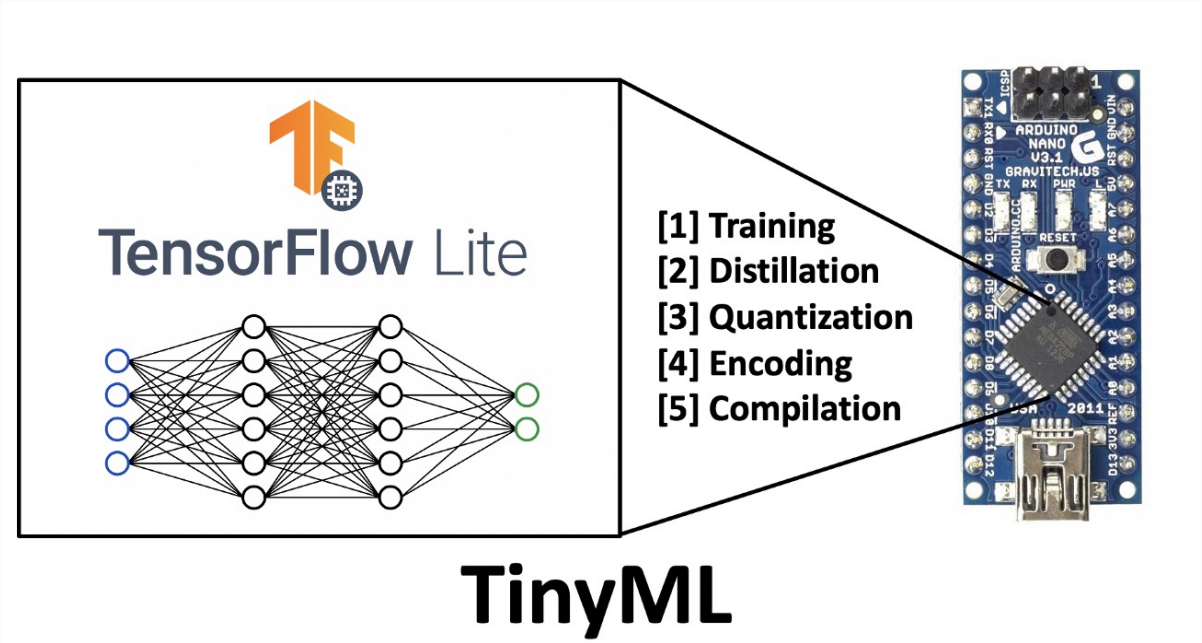
- Consists of two main components:
  - The **TFLite converter**, which converts TensorFlow models into an efficient form for use by the interpreter and can introduce optimizations to improve binary size and performance.
  - The **TFLite interpreter** runs with specially optimized models on many different hardware types, including mobile phones, embedded Linux devices, and microcontrollers.





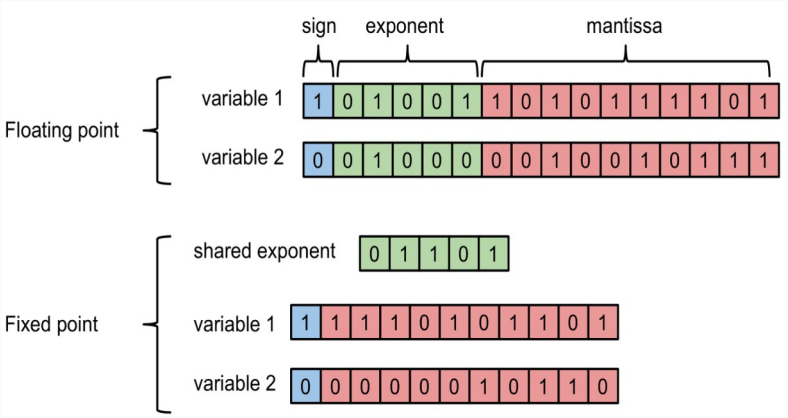
TensorFlow Lite optimizations

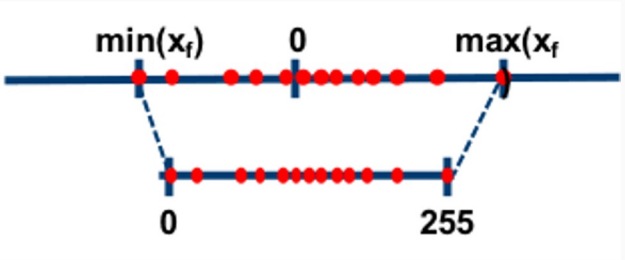
- Some optimizations:
  - Quantization
  - Pruning
- ..and
  - Separable Convolutions
  - Knowledge Distillation



Quantization

- Neural Networks generally use 32 bit floating point weights
- These require more memory to save and custom logic to execute efficiently (which many micro-controllers do not have)
- By converting weights to INT8 (8 bits), we can save memory and compute faster
- However, it leads to a loss in accuracy since we lose precision





source: heartbeat.fritz.ai

Ethics

MACHINE LEARNING SENSORS

Pete Warden [1] **Matthew Stewart** [2] **Brian Plancher** [2] **Colby Banbury** [2] **Shvetank Prakash** [2] **Emma Chen** [2]
Zain Asgar [1] **Sachin Katti** [1] **Vijay Janapa Reddi** [2]

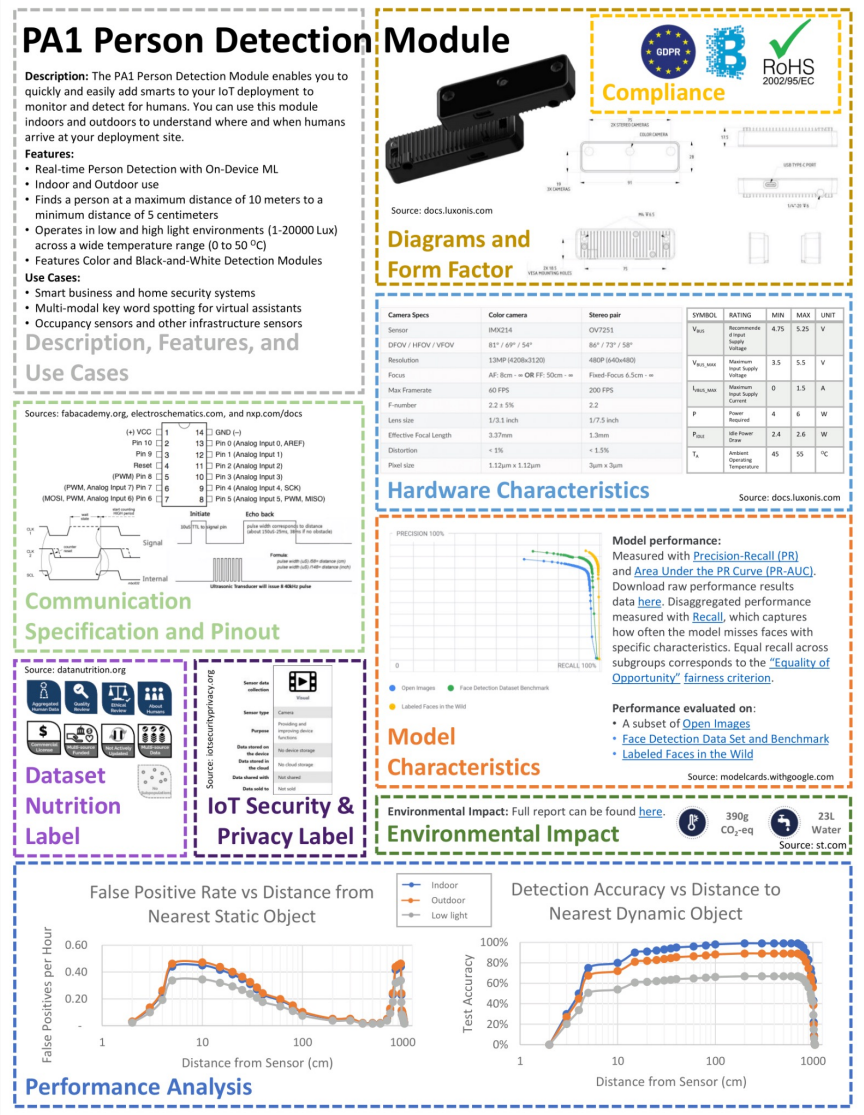
[1]Stanford University [2]Harvard University

ABSTRACT

Machine learning sensors represent a paradigm shift for the future of embedded machine learning applications. Current instantiations of embedded machine learning (ML) suffer from complex integration, lack of modularity, and privacy and security concerns from data movement. This article proposes a more data-centric paradigm for embedding sensor intelligence on edge devices to combat these challenges. Our vision for "sensor 2.0" entails segregating sensor input data and ML processing from the wider system at the hardware level and providing a thin interface that mimics traditional sensors in functionality. This separation leads to a modular and easy-to-use ML sensor device. We discuss challenges presented by the standard approach of building ML processing into the software stack of the controlling microprocessor on an embedded system and how the modularity of ML sensors alleviates these problems. ML sensors increase privacy and accuracy while making it easier for system builders to integrate ML into their products as a simple component. We provide examples of prospective ML sensors and an illustrative datasheet as a demonstration and hope that this will build a dialogue to progress us towards sensor 2.0.

https://arxiv.org/pdf/2206.03266.pdf

Ethics



https://mlsensors.org/

Future

TinyML will become the largest driver of the microcontroller market in the next 10 years, according to Remi El-Ouazzane, the president of STMicroelectronics' microcontrollers and digital ICs group.

"I really believe this is the beginning of a tsunami wave," he told EE Times in an exclusive interview. "We're going to see a tsunami of products coming with ML functionality: It's only going to increase, and it's going to attract a lot of attention."

STMicro has roughly a quarter of the microcontroller (MCU) market today, shipping between five and 10 million STM32 MCUs every day. According to El-Ouazzane, over the next five years, 500 million of those MCUs will be running some form of tinyML or AI workloads.

TinyML, which refers to running AI or machine learning inference on otherwise generic MCUs, "will become the largest endpoint market in the world," he said.

Key takeaways



TinyML = IoT + ML

Machine learning inference runs on the sensor device itself, at under 1 mW.



Process data at the edge

Local inference cuts latency and bandwidth, keeps data private, and leaves no good data behind.



Cheap hardware works

Low-cost microcontrollers like the ESP32 or Arduino Nano 33 BLE Sense (256 KB RAM) run useful models.



Train big, deploy tiny

Train in the cloud, then shrink with TensorFlow Lite Micro: INT8 quantization and pruning cut memory and speed up inference.



Applications are broad

Keyword spotting, vision with FOMO, anomaly detection, ECG monitoring, crop disease and wildlife counting.

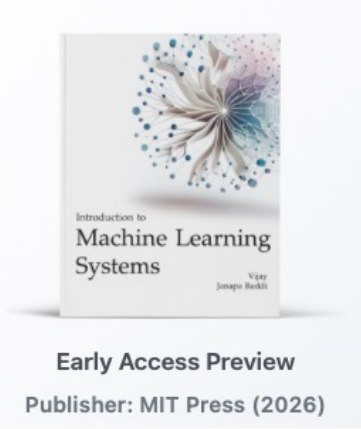


Grow responsibly

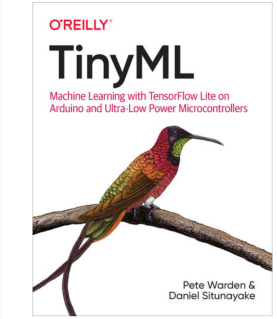
STMicro expects 500M of its MCUs to run TinyML within five years; ethics must be designed in.

On-line references

- Main refs
  - https://www.mlsysbook.ai

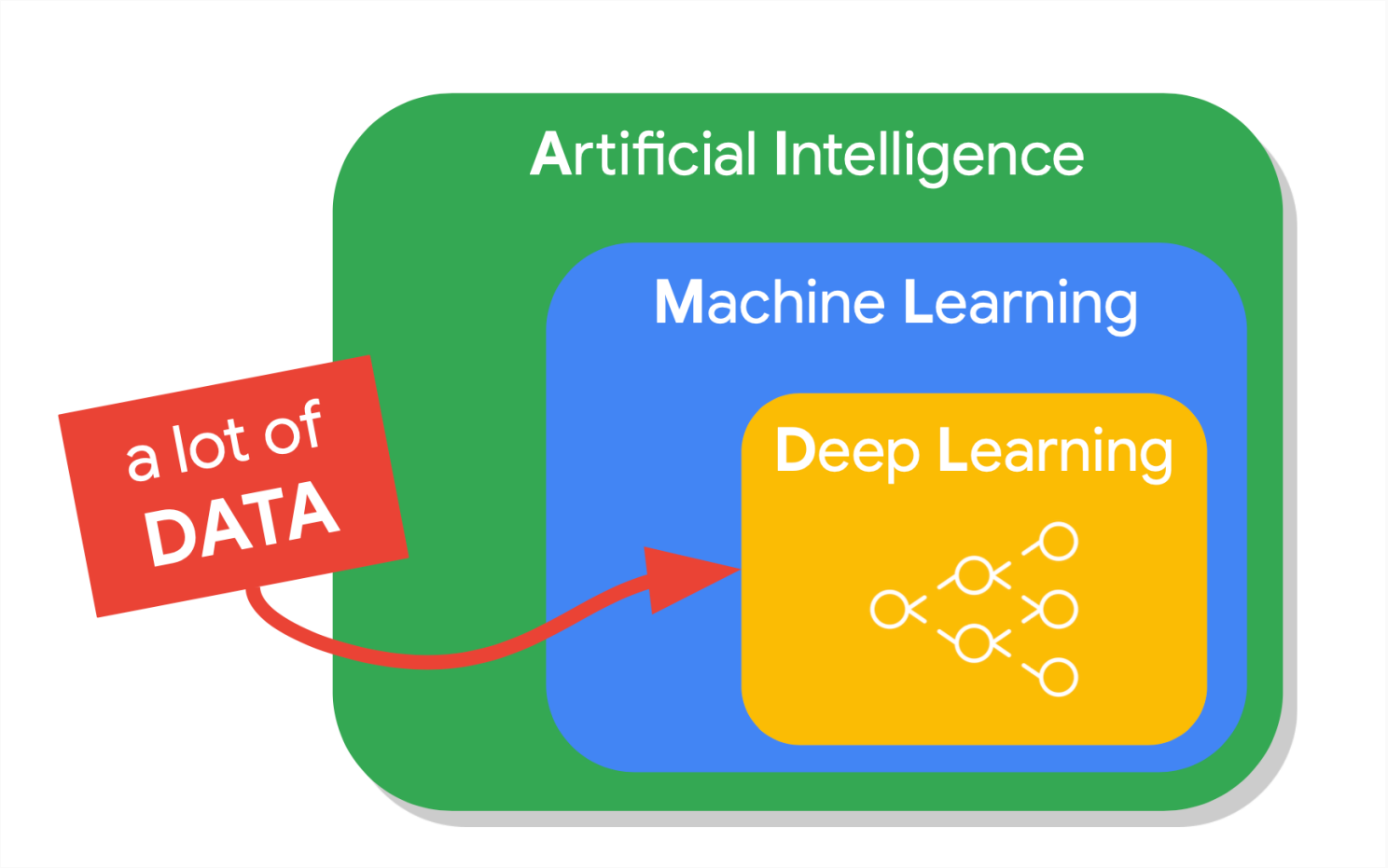


- TinyML Academic Community
  - http://tinymledu.org
- HarvardX Profession Certificate in Tiny Machine Learning (TinyML)
  - https://github.com/tinyMLx/courseware

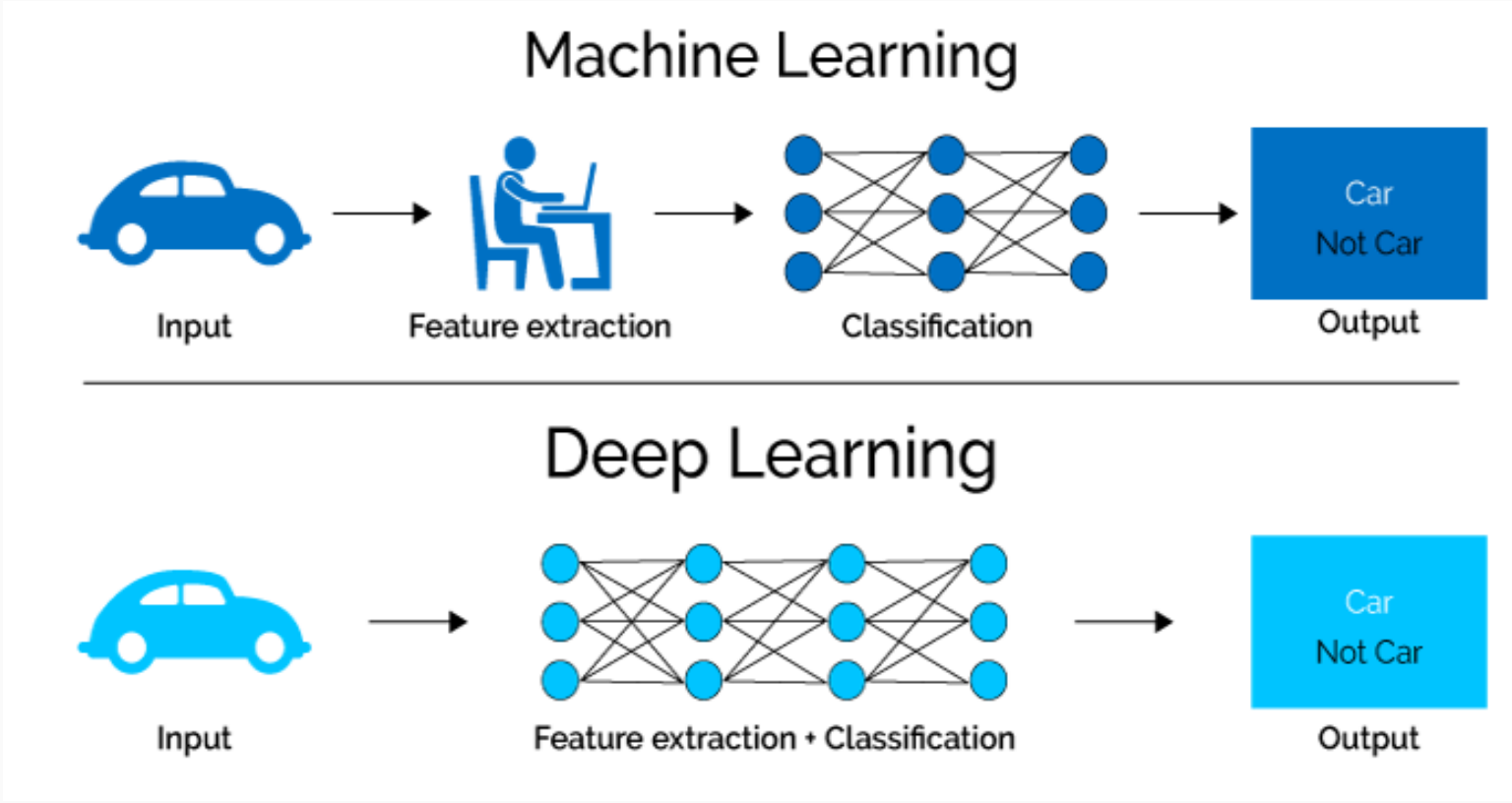


- Tensorflow lite:
  - https://www.tensorflow.org/lite/
  - https://experiments.withgoogle.com/collection/tfliteformicrocontrollers
- Seeedstudio: Everything About TinyML – Basics, Courses, Projects & More!

Machine Learning: Some initial definitions

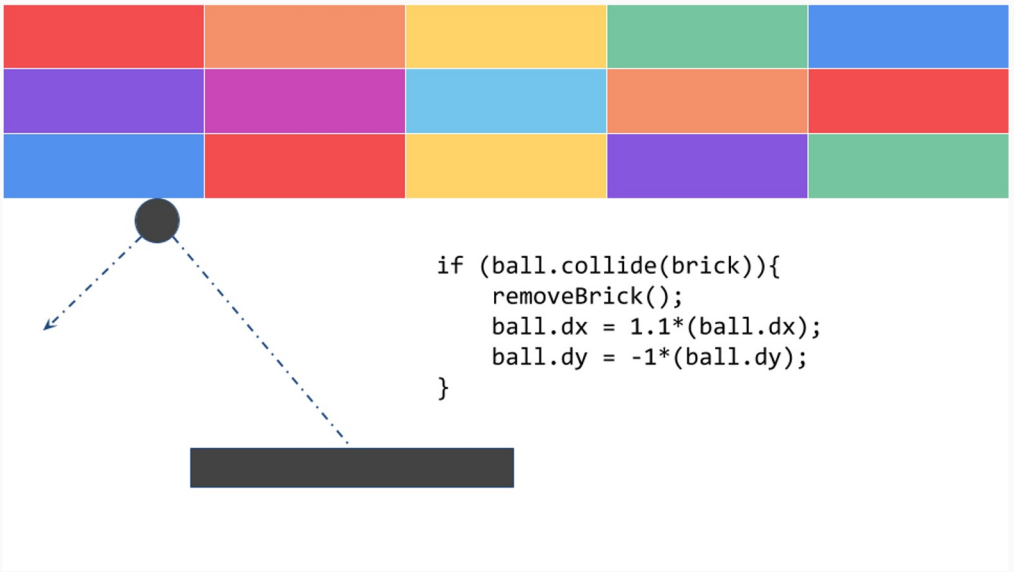


Machine Learning vs Deep Learning



Explicit Coding

- **Defining rules** that determine behavior of a program
- Everything is **pre-calculated and pre-determined** by the programmer
- **Scenarios are limited** by program complexity



The Traditional Programming Paradigm



Consider Activity Detection



```
if(speed<4){
    status=WALKING;
}
```



```
if(speed<4){
    status=WALKING;
} else {
    status=RUNNING;
}
```



```
if(speed<4){
    status=WALKING;
} else if(speed<12){
    status=RUNNING;
} else {
    status=BIKING;
}
```



```
// ???
```

Way too complex to code!

The Machine Learning Paradigm



Activity Detection with Machine Learning



```
01010010101001010101
10010101010010111101
01001010100101010010
01010010101001010101
```

Label = WALKING



```
10101001010010101010
01010100100100100001
00100111110101011111
10101001001111010111
```

Label = RUNNING



```
10010100111110101010
11010101110101011100
10101011110101010110
11111100011110101010
```

Label = BIKING



```
11111111110100111010
00111111010111110101
01011101010101011100
10101010101001111100
```

Label = GOLFING

Activity Detection with Machine Learning



```
01010010101001010101
10010101010010111101
01001010100101001001
01010010101001010101
```

Label = WALKING



```
10101001010010101010
01010100100100100001
00100111110101001111
10101001001111010111
```

Label = RUNNING



```
10010100111110101010
11010101110101011110
10101011110101001011
11111100011110101010
```

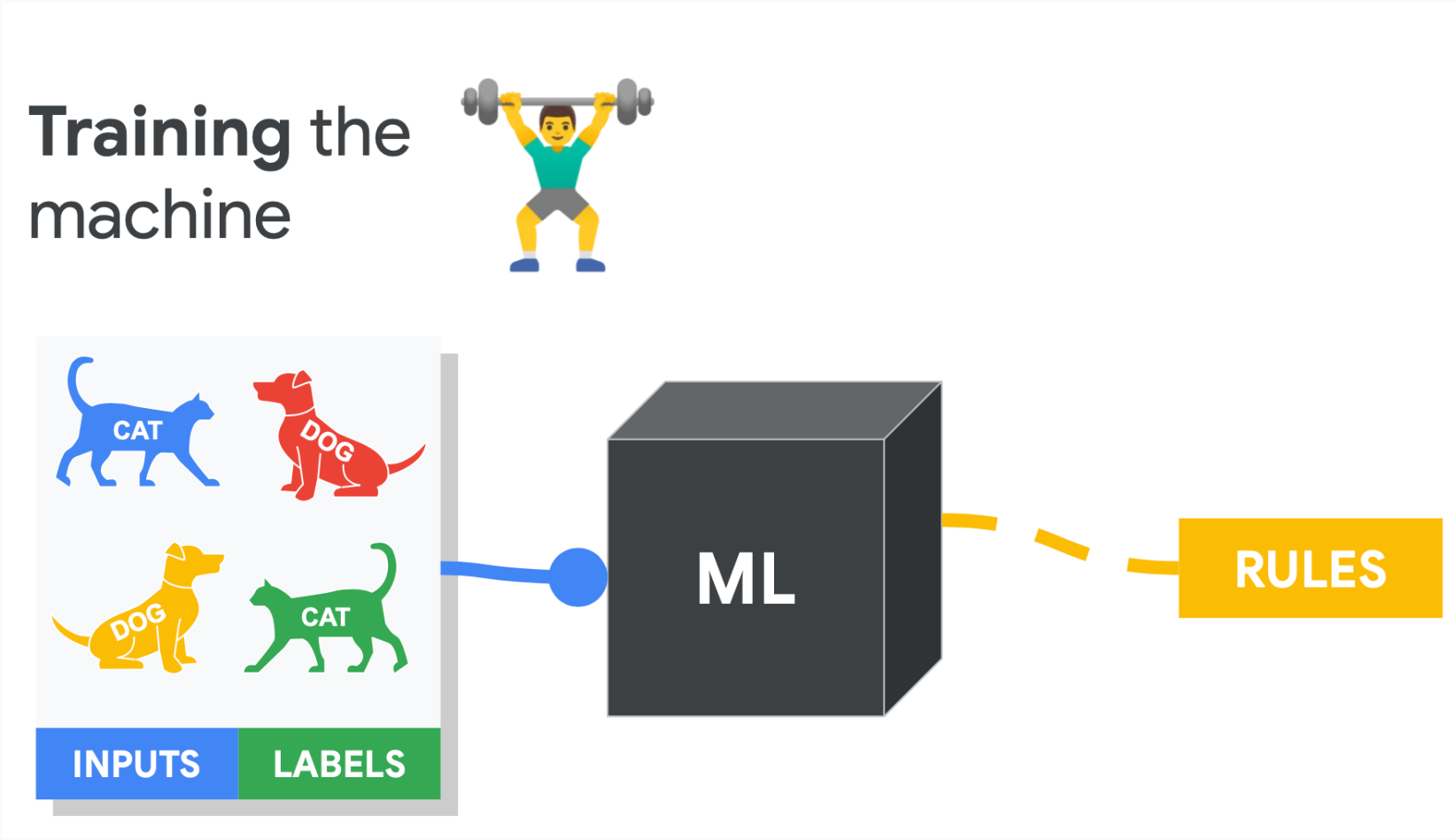
Label = BIKING



```
11111111110100111010
00111111010111110101
01011101010101001110
10101010101001111110
```

Label = GOLFING

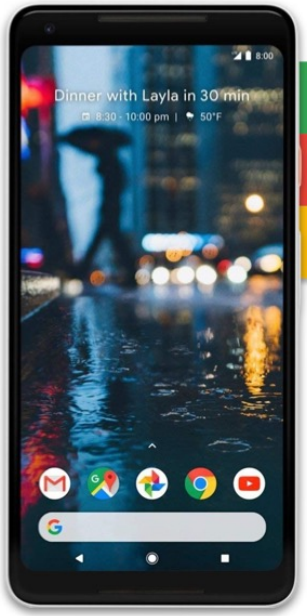
Training



Deploying



Battery & Memory



Less memory

Less compute power

Only focused on *inference*

Collect & Transform Data → Design & Train a Model → **Deploy Model** → Make Inferences

Predictions / Inferencing

